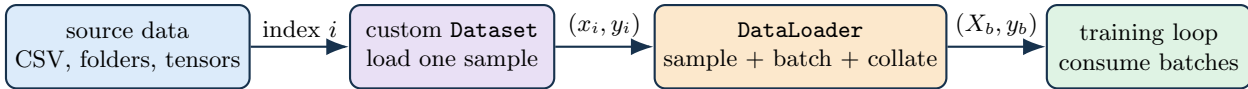
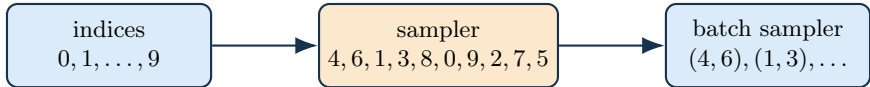


1 update/epoch
largest memory footprint

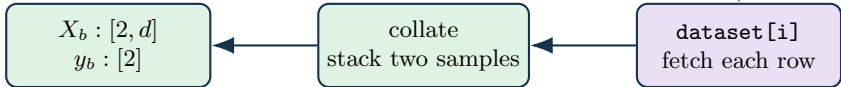
N noisy updates/epoch

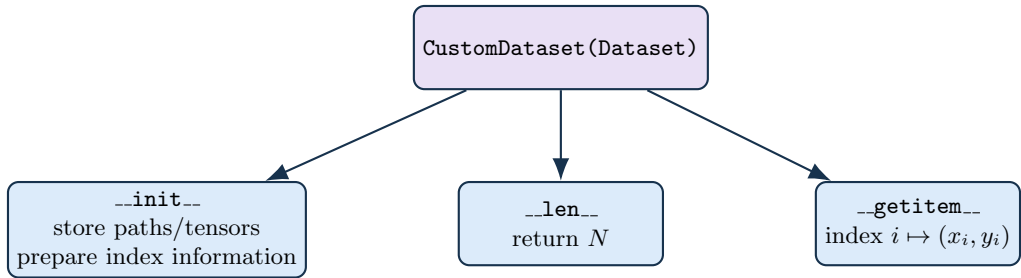
K efficient updates/epoch
practical compromise



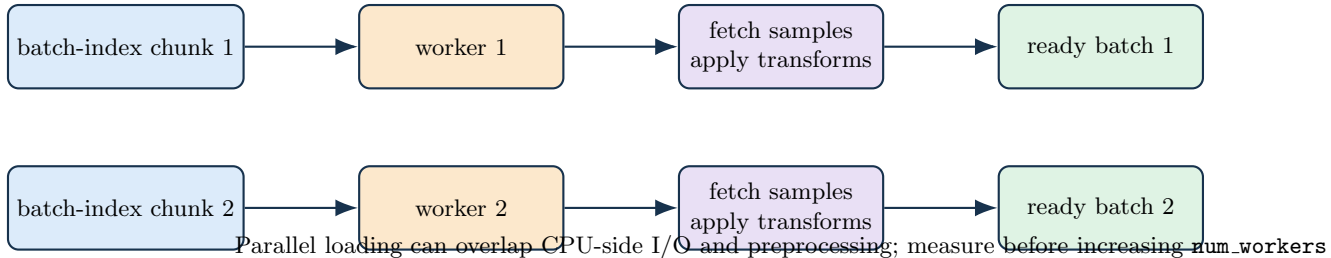


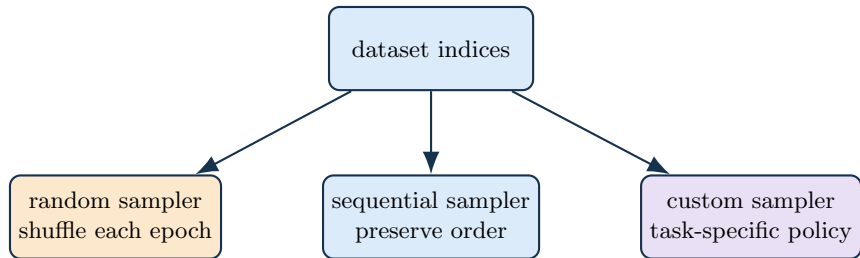
With $N = 10$ and $B = 2$, five batches cover one epoch





The methods support `len(dataset)`, `dataset[i]`, and `DataLoader` iteration



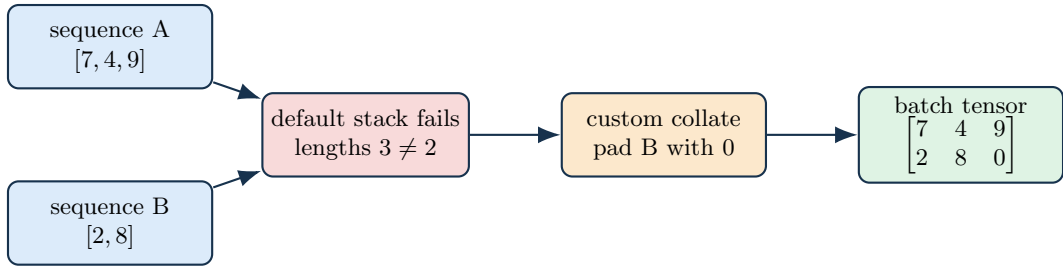


typical training

time series/evaluation

rare-class control
or weighted draws

At 99:1 imbalance, naive random batches may omit the rare class



Padding equalizes shape; masks or lengths should identify real tokens

